

DRAFT PERFORMANCE WORK STATEMENT

Software development support for NWC Joint Military Operations Department War at Sea Wargame.

1.0 Background

As a part of the Naval War College resident academic program, the Joint Military Operations (JMO) Department uses educational wargames to enhance student learning. Our primary gaming platform is *War at Sea* (WaS), a two-sided, double-blind wargame that allows teams of student players to apply course material in a simulated maritime combat environment by making military decisions to achieve operational objectives against a thinking enemy.

WaS uses large format tabletop game maps and cups of dice to represent forces and the combat environment. Game play involves a series of game turns, each representing a 12-hour period, where students first plan their moves on the game map and then communicate their intended actions to a faculty facilitation team for adjudication and resolution of tactical engagements.

Currently, students communicate game inputs to the facilitation team by drawing a map-graphic of their intentions on one or more plexiglass game cards. The facilitation team then adjudicates by overlaying the two teams' plexiglass game cards and manually inspects them for interactions between opposing forces.

This manual game card creation and adjudication process introduces opportunities for error and slows down game play. Digitization of this part of the wargame, i.e. digitizing the game cards and enabling adjudication on a tablet, would reduce errors and speed up game play, improve accuracy and reduce lost time, thereby enhancing the learning value of the game.

Note: WaS game play involves several other manual processes, including tactical combat resolution, imposition of capability loss due to damage, and logistics accounting. These other functions will not be digitized. These tactile interactions are important to achieving the game's educational objectives and will continue to be conducted manually.

1.1 Place of Performance: Contractor's facility and at the U.S. Naval War College, Newport, RI.

2.0 Scope

The contractor shall provide software development support services for the JMO department under the Dean of Academic Affairs. Specifically, this support is the development of an unclassified tablet application (app) to facilitate digital adjudication of WaS educational wargames.

3.0 Applicable Documents

Number	Title
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3.1	NIST Special Publication 800-37 Risk Management Framework for Information Systems and Organizations - A System Life Cycle Approach for Security and Privacy
3.2	DoDI 8500.01 Cybersecurity
3.3	DoDI 8510.01 Risk Management Framework (RMF) for DoD Information Technology (IT)
3.4	IEEE 829: Standard for Software and System Test Documentation

4.0 Technical Requirements

In accordance with the Applicable Documents (ADs) in Section 3.0 and using Government Furnished Information (GFI) in Section 6.0, the contractor shall provide software development support for the JMO Department of the Naval War College (NWC) by developing a tablet app to replace the manual game card creation and adjudication process of the WaS educational wargame. The app will allow players to submit their game inputs (movement, ISR and fires) and facilitators to adjudicate force interactions all on connected tablets.,

4.1 Application Development

- The contractor will develop and deliver a tablet app subject to the specifications herein.
- The contractor will consult with JMO subject matter experts throughout the development process and provide JMO with no less than two opportunities to play-test the app in each of the three game scenarios to evaluate functionality and provide guidance to enhance functionality and playability. The contractor will incorporate JMO feedback and resolve any issues exposed during playtesting.
- The contractor will coordinate with JMO to play-test the app on-site at the NWC with JMO provided personnel a minimum of three (3) times during development.
- The contractor will load system software, configure system settings, apply security and system related patches, in accordance with current Security Technical Implementation Guides (STIG) and applicable documents listed in Section 3.0. The contractor shall submit STIG reports upon completion of the application.
- The contractor will provide maintenance and support for the app for a period of 5 years after delivery including any patches or updates needed to retain full functionality.

4.1.1 App general characteristics:

- Platform. The app will run on a MS Surface Pro or equivalent Windows tablet and support input via a Surface Slim pen or equivalent stylus. There will be provisions for follow on support, as needed, to update the app to be compatible with future Windows releases.
- Game scenarios. The app will support the three current game scenarios with provisions for additional future scenarios. The current WaS scenarios are: a 1944 Pacific War historical scenario ("Leyte"), a 1982 Falklands-Malvinas War historical scenario ("F/M"), and a present-day fictional scenario set in the western Pacific ("Borneo").

- Game rule sets. The app will employ *War at Sea* rule sets for force movement and for detection (ISR). These rulesets are contained in the *NWC 2204 WaS Instruction Book*, and the scenario-specific *Orders and Forces* documents. Current versions of these documents will be provided by the JMO Department. The app will have provisions for the JMO Department to be able to make future minor changes to the rule sets.
- Classification. *War at Sea* is an unclassified game and involves no classified CUI or otherwise sensitive information. The app will be unclassified and will run on unclassified networks.

4.1.2 App general functionality:

- The app will support multi-factor authentication of users in the Azure environment. Users will be granted access by name to the app by the College administrators as either players (students) or facilitator (faculty).
- The app will support running up to 10 stand-alone games simultaneously. All simultaneous stand-alone games will have full functionality, but there will be no interaction between games, only between players and facilitators within the same game.
- The app will support simultaneous use of up to nine tablets within each game: six player tablets (three per team) and three facilitator tablets.
- During game play, tablets within each stand-alone game will connect to the server via WiFi. Data transfer from the server will connect tablets within each game and will include the transmission of player movement, ISR and fires inputs (i.e. digital version of current physical game cards) to facilitators for adjudication, and transmission of force end-of-turn positions from facilitators back to players. Note: all gaming will occur in locations where WiFi is available.
- The app display will mirror the appropriate scenario game map, will be pan and zoom capable, and will display suitable graphics for all forces, including land, maritime, air, missiles and minefields as well as neutral (white) shipping where appropriate to the scenario. The app will display both force positions and path of movement (tracks). Tracks will be identifiable as to the specific force they represent by either a persistent or pop-up label. Force icon labels will mirror existing force titles. Graphics for each team will be color-coded, nominally blue for the blue team and black for the red team.
- All player cards (move, ISR, deliberate fires) will be labeled with a facilitator-generated unique game identifier (i.e., *ILC 2025 Leyte game #3*), the team (red or blue), the turn number, whether it is a day or night turn, and the card type.
- The user interface will employ contemporary tablet user interface best practices to the greatest degree possible to minimize the learning-curve overhead. It will be simple, menu driven, primarily graphical, and support the use of a stylus for input.
- The app will support the following modes: setup, run, pause, replay (single or all tablets) and end game. Modes will be selectable only by facilitators.
 - Setup mode- Setup mode will allow facilitators to initiate a new game, to assign students to teams, and to add or delete forces to the scenario-specific force list. Setup mode functionality will be accessible only by facilitators.

- Run mode- Run mode will enable players to create and submit inputs (movement, ISR and fires cards), and facilitators to adjudicate player inputs for trip-overs, deliberate fires and detections. Run mode will also enable facilitator admin tasks such as accept player cards, return cards to players for editing, and advance turn counter to next turn.
- Pause mode. Pause mode will disable input and editing and save all data for an indefinite period.
- Replay mode- The app will save all game play data for retrieval to support a post-game after-action review. Replay mode will have two sub-modes:
 - Replay single tablet. Accessible by facilitators during run mode, puts a single facilitator tablet into replay mode without change to all other tablets. This function allows one facilitator tablet to be used to retrieve and export non-editable versions of all movement, ISR, and fires cards, and all facilitator-accepted detections, deliberate fires and trip-overs up to that point in the game. The purpose of this functionality is to support the development of a running after-action document for use after the game ends.
 - Replay all tablets. Accessed at the conclusion of the last turn of a game, This function allows all tablets to be used to retrieve and export non-editable versions of all movement, ISR, and fires cards, and all facilitator-accepted detections, deliberate fires and trip-overs for the whole game. The purpose of this functionality is to allow players and facilitators to review game details during the post-game after-action review.
 - Replay archived/saved game. The app will support retrieval and replay of non-editable versions of all movement, ISR, and fires cards, and all facilitator-accepted detections, deliberate fires and trip-overs from a previous game. All games will be saved/archives at end game.
- End game. Saves all game data, ends game play.

4.1.3 Player functionality.

The app will replicate as closely as possible the function and appearance of the current manual game card system while speeding up the process and minimizing opportunities for errors. Players will be admitted to a particular stand-alone game and to red or blue team within that game by the app upon login based on player game/team assignments made by the facilitator during game setup. See Player tablet process flow chart below.

- Each team (red, blue) will have three player tablets. Any player tablet on a team will be able to display and edit either the move, ISR or deliberate fires card, but only one tablet at a time, per team, will be able to show any card. In other words, once one of the player tablets on a given team has selected the move card for editing, the other two player tablets on that team will only be able to select either ISR or deliberate fires. All turns will involve a movement card, while ISR and deliberate fires cards are scenario and situation dependent.

- When a new game is first put into run mode (by facilitators), the app will enable player inputs and will present the move card player with a menu of their team's maritime, land, and land-based air forces* for initial positioning. These will include all forces from the appropriate *Objectives and Forces* document as modified by the facilitators during game setup, should they choose to do so. The move card player will be able to select and drag each force to an initial starting position which will populate on that team's ISR and fires tablets. The app will not impose constraints on initial force positions. Facilitators will impose appropriate constraints offline.

*Carrier-based air forces will not be given an initial position, but rather will be selected from a menu as needed for input to ISR and deliberate fires cards. This is because, unlike land-based air forces, they are modeled in the game as capability (dice) within the maritime carrier force (dice cup) rather than as a discrete force. This applies unless players choose to recover a carrier-based asset on land, at which point it becomes a land-based force and will be displayed as such.

- Starting positions of all friendly forces will be displayable on all player tablets at the beginning of each turn and players will be able to select which forces are displayed, maritime, air, land in any combination. For the first turn of a game, these start positions will be entered by the move card player as explained above. For subsequent turns the start positions of all forces will be the location where that force ended the previous turn, transmitted to player tablets from a facilitator tablet when the facilitator selects "next turn" at the end of the adjudication process.
- Player tablets will be enabled for input when: (1) facilitators initially put the game in run mode, and (2) facilitators select "advance to next turn" during game play. When enabled for input, player cards are in "draft" mode. When the players determine that a card is complete, they will be able to "submit," which transmits that card to the facilitators and changes its status to "submitted," making the card viewable but no longer editable. Facilitators will either accept the card, or "return for editing," resetting a card to "draft" status.
- Player input will differ between the three player tablets as follows:
 - The move card player will input intended tracks of all maritime forces, land forces, land-based air forces and the establishment of minefields. Positions of maritime forces, land forces, and land-based air forces will always be displayed on the move card even if a force is stationary. Carrier-based air forces will only display when moving. If there are any forces newly arriving during a turn, the move card player will be able to select those forces from a menu and drag it to an initial starting position before making any card inputs. the move card player will be able to create any new disaggregated forces allowed by the game ruleset. The move card player can also delete any forces lost during combat in the previous turn.
 - The ISR card player will be able to input the tracks of all air forces dedicated to ISR missions for that turn.
 - The deliberate fires card player will input the tracks of any air forces and missiles dedicated to deliberate fires missions for that turn.

- Functionality for player inputs:
 - Maritime, land forces, and land-based air forces. Players will be able to draw a route of movement by selecting and dragging a force icon from its starting position (either initial starting position or ending position from previous turn) along the intended route to the intended end position.
 - When inputting submarine movement, the player will be prompted to select surface or submerged, and the icon for that force will reflect its status.
 - Multiple surface forces can be grouped and moved along a single route. When this is done, the player will be prompted to select “touching” or “stacked” and the force icon will reflect this status.
 - Where appropriate, the player will be able to select and the app will indicate CAP being flown over a force.
 - Carrier-based air forces. Players will be able to select forces from a menu and drag to a starting point at a suitable air-capable maritime force, and from there draw a route of movement by dragging along the intended route to the intended end position limited to the maximum range allowed. The app will not need to determine whether or not that air capability is actually available (i.e. whether there are blue or yellow dice in the cup). Facilitators will do this manually offline.
 - Force routing. Players will have multiple options for drawing routes, including freeform, direct point-to-point or via waypoints. Players will be able to delete or edit routes until the card is submitted. Editing functionality will include “rubber-banding,” i.e., touch-hold-drag a point on the route line to establish a new waypoint. Depending on the scenario, air routes may also include one or more menu-selectable options including ISR search rings (6 or 9 inch), Combat Air Patrol (CAP) stations, airborne refueling station, or Airborne Early Warning and Control (AWACS) station as appropriate to the scenario and force chosen. Air routes will display the number and type of air forces (dice color) assigned (i.e. 3B, 2Y, etc.)
 - Tethered CAP station. Where allowed in the scenario, CAP stations may be tethered to a maritime force, meaning the CAP station moves at a fixed distance and bearing from the maritime force.
 - User-defined force conditions. For forces where either the movement or detection rule set is contingent upon a user-defined condition, the player will be prompted to toggle that condition on or off and that status will be indicated on the affected force icon. Examples include Emissions Control (EMCON), submarines surfaced or submerged, and Maritime Patrol and Reconnaissance Aircraft (MPRA) (yellow dice) carrying anti-submarine detection capability (purple dice).
 - Range limitations. For all movements, the app will apply the appropriate WaS rule set to limit the route to the maximum range of that force in its current status (i.e., sub surfaced or submerged, surface force moving fast or slow, or in littoral waters, etc.) If a player attempts to draw a route more than the allowable range,

the app will truncate the route at the farthest-on point allowable under the rules. In cases involving air-to-air refueling (AAR), the app will apply the appropriate range rule sets for both the refueling aircraft (AAR tile) and the tactical aircraft it is supporting.

- At the end of a game, i.e., when the facilitators select “replay for all tablets”, players of both teams will be able to retrieve, display and export move, ISR and fires cards of both teams from all turns, including all trip-overs, deliberate fires and detections.

4.1.4 Facilitator functionality:

In general, the app will present facilitators with a highly tailorable and easy to interpret graphical representation of the player’s intentions to enable them to determine where and how opposing forces interact. See facilitator tablet process flow chart below.

- There will be three facilitator tablets in each game. All facilitator tablets will have the same functionality. Normally during game play, two of the three facilitator tablets will be in run mode supporting the faculty members adjudicating force interactions while the third facilitator table is in replay (single tablet) supporting the student assessment team.
- Upon initial login, facilitators will be able to select either a new or existing game. Choosing new game will put the tablet into setup mode where facilitators will have the necessary administrative control to title a new game, enroll students by name into blue & red teams and assessor roles, and modify force lists by adding or deleting forces from either team.
- During game play (run mode) facilitator tablets will receive game cards from each player team via the app when players “submit” that card which transmits that card to the facilitators and changes its status from “draft” to “submitted,” making the card viewable but no longer editable on the student tablet. Facilitators will be prompted to accept or “return for editing.” Facilitators will also be able to edit cards themselves rather than returning for edit to the players.
- Once accepted, the app will overlay cards from opposing teams (in any desired combination) and highlight the locations of all potential interactions between opposing forces for the facilitators to evaluate. In determining potential interactions, the app will apply the appropriate WaS rule-sets for movement and detection.

Note: The app will identify all interactions that are possible under the movement and detection rules and will not apply the probability rule sets (i.e., dice rolls). Facilitators will apply probability rules manually, offline.

- Facilitators will have maximum flexibility in choosing how the app displays potential interactions. Facilitators will be able to select to see all potential interactions simultaneously or to see individual potential interactions sequentially beginning with the first along the route of a selected force. When zoomed in to inspect complex interactions, facilitators will be able to select individual tracks and have the app display all relevant force information for that track such as, force type and quantity, status (EMCON, moving fast/slow, submerged, etc.) Facilitators will be able to toggle on or off any combination of the following.
 - Blue team movement card

- Red team movement card
 - Blue team ISR card
 - Red team ISR card
 - Blue team fires card
 - Red team fires card
 - Only air forces
 - Only maritime forces
 - Only forces transmitting (that is, not in EMCON)
 - Trip-overs
 - Deliberate fires
 - Detections
- For each potential interaction identified, the app will prompt the facilitator to accept or ignore. If accepted, the app will further prompt for designation as either a trip-over, detection, or deliberate fire.
 - If an interaction is accepted and designated by the facilitator as either a trip-over or a deliberate fire, the app will assign it a number in sequence (i.e., TO#1, TO#2, DF#1, DF#2 etc.) and indicate the number at that location on the display. If an interaction is accepted and designated as a detection, no number is assigned, and movement of the detecting force does not stop. The app does, however, mark, highlight, or otherwise indicate that the force has been detected.
 - Facilitators will be able to annotate freehand notes on each of the cards while evaluating them for adjudication for use while briefing back to the student teams.
 - For land force trip-overs or deliberate fires, force movement stops at that point. For maritime force trip-overs or deliberate fires involving a single force or stacked forces, movement stops at that point. For multiple maritime forces not stacked, the facilitator chooses which stop and which continue. For trip-overs involving air forces, the facilitator will be able to choose whether or not the force continues.
 - The app will not have any role in communicating the results of facilitator adjudication back to the players. Facilitators will brief-back trip-overs, detections, and deliberate fires manually (offline), and will manually take trip-overs and deliberate fires to the battlebox for combat resolution. The app will allow facilitators to selectively display accepted interactions (trip-overs, deliberate fires and detections) in any sequence or combination for easy reference while briefing players.
 - After manual combat resolution, facilitators will be able to delete any forces killed at the battlebox.
 - Before selecting “advance to next turn,” which returns ending positions to players, facilitators will be able to alter the ending position of any force from that determined by the app.

5.0 Status Reporting

Monthly Reports

The Contractor shall prepare and provide monthly progress reports to include schedule, technical accomplishments, and risks. These reports are due on the 15th of each month.

6.0 Government Furnished Information (GFI)

The following Government Furnished Information (GFI) will be made available under this contract:

Number	Title
6.1	War at Sea Documentation (available upon request)
6.2	WaS tablet app process_Facilitator tablet flow pages 1 and 2 (below)
6.3	WaS tablet app process flow_Player tablets (below)

7.0 Software Development Plan

Contractor shall provide a complete development plan to develop the application as a cloud-based service hosted in the MS Azure environment using Windows Server 2022 (current) and Windows 11 (current) for client-side requirements. This performance plan should include (at a minimum):

- Proposed application architecture
- Proposed application development language
- Proposed network/communications requirements and application operational requirements including server CPU, Memory, data storage requirements, and video support requirements.
- Proposed timeline and milestones for project development phases including:
 - Discovery
 - Detailed functionality
 - Project wireframe and workflow diagrams
 - Conceptual User interface for Government Approval
 - Development
 - User testing at each phase of development
 - Change management
 - Migration to production environment
 - Deployment and testing
 - Each phase of the project shall be reviewed with the NWC personnel (JMO & IRD) for acceptance and conformity to project goals.
 - Contractor should identify any impediments foreseen at any phase of the project to the NWC team in order to identify possible resolution collaboratively.
- One year of ongoing application support from the date of delivery and up to a four (4) year option to be executed based on need availability of appropriated funds.
 - Support shall include any changes required to continue compliance with NIST standards, changes due to computing environment updates, etc.
- Cost by development phase.
- Other considerations for the proposed application must include integration with existing systems, authentication methods, and user accessibility features. All development shall be

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in conformance with NIST Secure Software Development Framework (NIST800-218A) and include applicable controls compliant with NIST 800-53, NIST 800-171 to achieve compliance with DOD/DON standards. All data shall be encrypted at rest and in motion. No data shall be stored or transmitted in clear text.

- ALL equipment purchased by the vendor to successfully develop and deliver this application as part of this contract shall be considered government equipment and shall be serialized and labeled as such and shall be turned back over to the Government upon delivery of the application in production. All hardware and operating systems shall meet applicable DISA Secure Technical Implementation Guides (STIGs) and be configured as such for development and through deployment.
 - Tablets and/or Laptop devices shall be configured and managed by NWC Information Resources Department (IRD) and managed via MS Intune Mobile Device Manager (MDM) prior to and through development, in addition to post-delivery to the NWC.

8.0 Data Rights

The Government shall retain full ownership and Unlimited Rights to the application, associated code, and all data. Once delivered, the contractor has no rights whatsoever to reuse, sell, or redistribute the code or application developed under this contract without express written consent of the US Government.

9.0 Security Procedures and Base Access

Temporary contractor badges and vehicle decals/passes in accordance with the process described below are required for base access. The contractor is required to fill out and process a SECNAV Form 5512-NSNPT form and submit to the security at least 14 calendar days before crew arrival. US citizenship is required. Items needed to obtain badges and vehicle decals are as follows:

BADGES

- I-9 (See your company Human Resource Director)
- Real ID Driver's License or other identification (i.e., Social Security Card, Birth Certificate, or Passport) NO COPIES WILL BE ACCEPTED AT PASS OFFICE

VEHICLE PASS

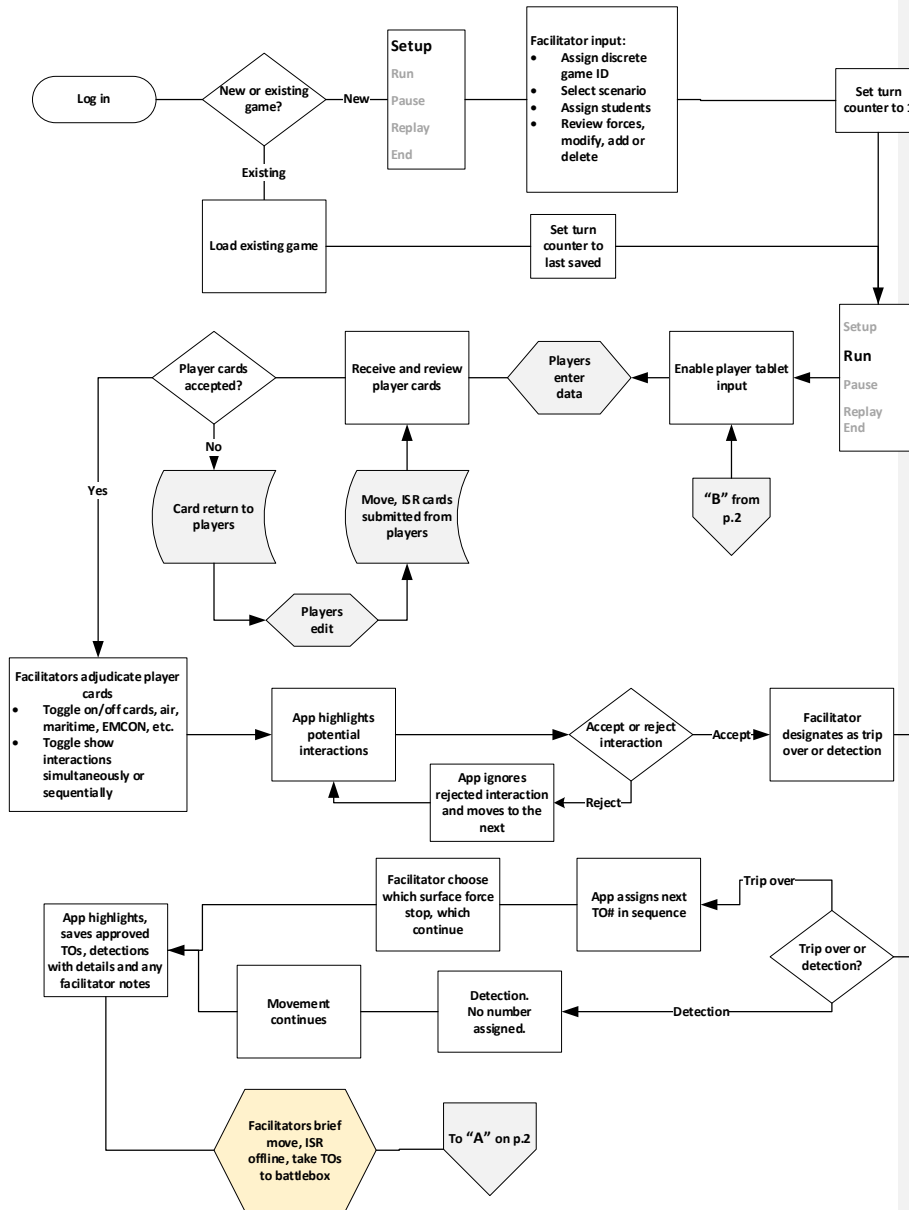
- Vehicle Registration
- Proof of Insurance
- Proof of Inspection

The contractor shall contact the government escort prior to arrival. The place of performance of all work is within secure facilities. The contractor personnel shall be required to remain with the government technical assistant/escort at all times during performance.

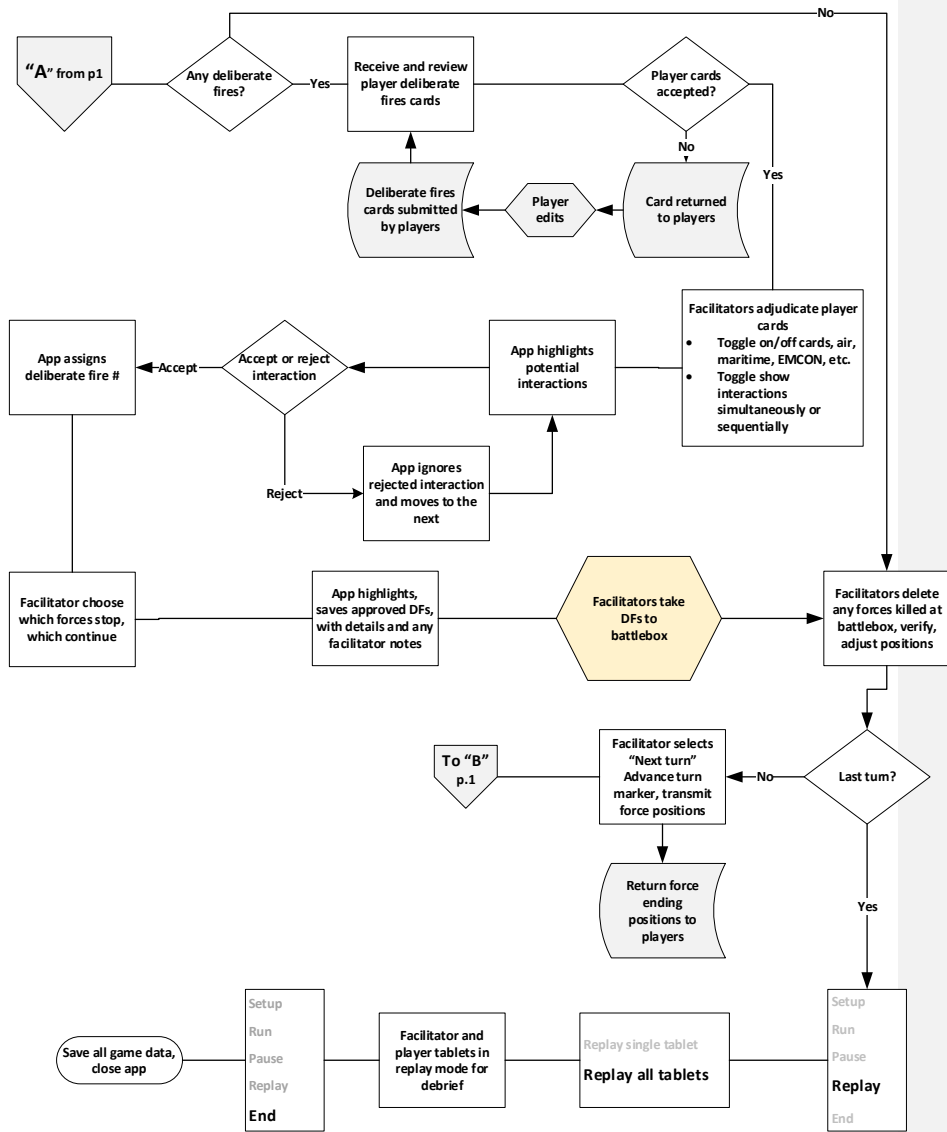
The contractor shall submit a valid visit request to Naval Station Security In-processing Center prior to performance of the task order at that facility. All contractors performing services at the Naval War College facilities shall be required to sign in and sign out at the Conolly Quarterdeck.

All contractor personnel shall be provided and must display the visitor's badge issued by the Security guards at all times.

WaS tablet app process
Facilitator tablet flow page 1



WaS tablet app process
Facilitator tablet flow page 2



WaS tablet app process flow Player tablets

